

Convolutional Neural Networking Based Automated Detection of Mango Leaf Diseases

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Abstract—Effective control of foliar diseases is crucial for mango trees to preserve their health and productivity. The system automatically diagnoses diseases that affect the leaves of mango plants. It detects early leaf symptoms by identifying indicators of common diseases, including anthracnose, through high-throughput screening of images captured via drones or stationary cameras. Diseases such as powdery mildew, bacterial infections, and black spots are detected using sophisticated computer vision and prediction techniques, including feature extraction and machine learning classification, to provide accurate diagnoses and faster treatment. Three primary criteria were the focus of our system in this study: (1) Image Processing Algorithms, (2) Machine Learning and Deep Learning Techniques, and (3) Leaf Disease Detection Procedures.

Keywords—Image Processing, Leaf Disease, Productivity, Mango Orchard, Disease Detection, Mango Leaf Disease, Tree Disease

I. INTRODUCTION

Agricultural industries form the backbone of many economies and are essential to ensure food security. Among the plants, mango tree is one of the most economically important fruit in terms of commercial value, it is widely grown in the tropics and subtropics but with different conditions. But the main challenge in mango tree production are diseases which affect crop yield and high quality. Traditional diagnostic methods rely on constant expert monitoring, which is time-consuming, difficult to work intensively, and can lead to human error.

Modern image processing and deep learning techniques have shown great potential for accurately identifying plant leaf diseases, offering real-time and automated diagnosis that can reduce dependency on manual inspections [1], [2].

This study aims to address these challenges faced by several organizations. We have developed an Image processing Mango Leaf Disease Detection system which provides us the following contributions

1. Users can be able to know by which disease the tree is

affected.

2. Automated Spray System will be able to spray medicines to the affected leaf.

3. Productivity of the mango will be increased as the disease detection will be easier with the help of this machine.

Convolutional Neural Networks (CNNs) and transfer learning approaches have been used effectively in similar research to classify plant leaf diseases, achieving high accuracy and robustness even with heterogeneous datasets [3]. In modern times, increasing image processing and machine learning have changed agricultural practices, providing effective automated answers to crop management and their disease detection. By integrating these technology with hardware automation, it's far possible to broaden clever systems capable of detecting disease and optimizing automated spray in a centered way. Such systems no longer most effective beautify productivity but additionally sell sustainable aid utilization.

This mission makes a specialty of the development of a complete device to detect mango leaf diseases using image processing strategies. Leveraging machine getting to know algorithms, the system pursuits to accurately identify diseases affecting mango tree leaves. To supplement this, hardware integration enables automated spray, making sure that only the affected regions get hold of the essential medicine. This twin capability addresses vital elements of agriculture: safeguarding crop health and protecting.

Image processing integration improves the overall effectiveness of disease management in mango orchards by enabling automated spraying, real-time monitoring, and decision support [4], [5].

II. RELATED WORK

Mango leaf disease detection is an important area of research aimed at enhancing agricultural productivity and safeguarding one of the most vital tropical fruits. The application of image processing and deep learning techniques has emerged

as a promising answer for accurate, efficient, and real-time disease detection. This thesis takes special advantage of the advances in knowledge acquisition algorithms in Convolutional Neural Networks (CNNs), turning to the challenges posed by heterogeneous datasets and complex real-world agricultural situations.

Studies have tested the performance of deep learning models in plant disease classification. For example, DenseNet-121 achieved an accuracy of 90.81% in detecting plant diseases, and demonstrated the potential of automatic prediction [6], [7] and techniques to identify problem in whole plants. Intensive studies evaluate these models capabilities and the deployment demands and situations are highlighted. [8], [9].

Addressing data set limitations has further articulated the superiority transfer learning approaches, emphasizing the need for meaningful, subjective data sets that enforce accuracy and better robustness of real-world systems [10], [11]. Besides mainly CNN-based methods, advanced plant disease pests. and have prestigious skills, where in various scenarios Despite the reliance on nice dataset and deployment difficulties [12], for example, corrective robots equipped with spectral sensors have had moderate success (66.4% accuracy) in an asymptomatic pathogen found in carrots [13].

New visualization techniques have once again shown promise in real field situations. A patented device successfully detected the signs and early symptoms of wheat diseases such as rust, septoria and tan spot with more than 80% efficiency by using a hierarchical algorithm pipeline [14], [15].

Challenges of dataset heterogeneity and real-time variability continue to be sizable. However, the capability of deep learning to enhance classification accuracy and advanced imaging techniques in realistic applications has been documented. [16], [17].

Overall, The field continues to make strides in resolving dataset barriers, enhancing early detection, and growing the robustness of models for actual-world applications in agriculture [2], [18]. the research community also emphasizes addressing emerging threats, consisting of the latest discovery of *Aureobasidium melanogenum* fungus, which poses a critical risk to mango production in a few locations [19].

III. PROBLEM STATEMENT

Mango Tree farming faces significant challenges because of the prevalence of disease along with Anthracnose, Bacterial Canker, and Powdery Mildew, which affect crop yield and quality. Traditional strategies of ailment detection depend upon manual inspections, that are time-consuming, labor-intensive, and prone to human error, in particular in massive-scale agricultural setups.

Although advances in artificial intelligence, especially convolutional neural networks (CNNs) have shown promise in addressing various agricultural demand scenarios, there are still a number of obstacles e.g. Gated-Recurrent Convolutional Neural Network (G-RecConNN) has 95% accuracy in early diagnosis those regarding dataset range and computational

complexity Faced with demanding conditions, Optimal MobileNet CNN (OMNCNN) took the highest performance in classifying tomato leaf diseases with 98.7% accuracy, outperformed VGG16 and AlexNet by feature extraction and hyperparameter optimization but the version constant the need to improve real-time muscle performance for a number of control situations [20], [25].

Pre-trained CNN algorithms such as DenseNet-121, ResNet-50, and VGG16 divided data sets into 38 instructions, with DenseNet-121 having an accuracy of 90.81%, indicating that the power of perform variational analysis but the effectiveness of their styles depends closely on the selection and diversity of datasets [21].

Other studies highlighted differences between laboratory and real-world applications, where limitations in the relevant data-set are major barriers. Although CNNs proved to be accurate for automated crop diseases, real-world scenarios require the inclusion of different types of data to be effective [4], [23]. RobHortic Spectral imaging field robot carried out fairly low accuracy in detecting asymptomatic pathogens in carrots (66.4% in laboratories and 59.8% inside the subject), once more highlighting the want for models emphasize adaptation to real-world conditions [22], [26].

This project addresses those demanding situations by way of integrating superior photograph processing, device learning, and automated spray systems into an integrated system. By using fashions inclusive of MobileNetV2 for actual-time ailment detection and integration with focused spray systems, the solution seeks to improve agricultural efficiency and improve crop health beneath numerous, actual-world situations.

IV. METHODOLOGY

The method followed in this project focus on developing a real time-mango tree leaf disease detection and automated spray system, using a combination of image processing, machine learning, and hardware automation. The goal is to pick out diseases in mango leaves with excessive accuracy.

A. Data Collection and Preparation

The related work study provided insight into the existing methods and limitations of disease diagnosis and automated spray, highlighting the need for time itself explores a more integrated approach in Bangladesh which was found to lack emphasis on local mango varieties and organized agricultural practices. The sample data were collected from publicly available sources [27] and field-collected images of mango leaves. The dataset collected 8 categories of leaf images which contains Healthy leaf as well as disease type of anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Powdery Mildew, Sooty Mould. Some samples are shown in Fig 1. The pictures were resized to 240×320 pixels corresponding to Convolutional Neural Network (CNN) input requirements. Data enhancement techniques such as rotation, flipping and light adjustment were used to increase data diversity and increase model robustness.

TABLE I. COMPARATIVE ANALYSIS OF MODELS ALONG WITH ACCURACY

Reference	Model	Accuracy	Appliance
[20]	Gated-Recurrent Convolutional Neural Network (G-RecConNN)	95%	Early diagnosis of plant diseases
[20]	Optimal MobileNet CNN (OMNCNN)	98.7%	Tomato leaf disease classification with feature extraction and hyperparameter optimization
[1]	LeafNet	98.55%	Common mango leaf disease detection using MangoleafBD dataset of 3.25 Million Parameters
[21]	DenseNet-121	90.81%	Classification into 38 instructions for disease detection
[21]	ResNet-50, VGG16		Used for dataset classification and variational analysis
[22]	RobHortic Spectral Imaging Field Robot	66.4% (lab), 59.8% (field)	Detection of asymptomatic pathogens in carrots
[4], [23], [24]	Generic CNNs	Varies	Automated crop disease detection limitations in real-world applications

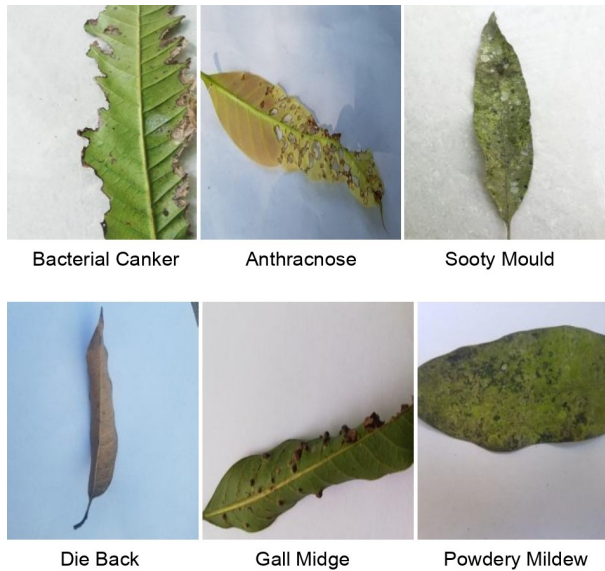


Fig. 1. Various types of Mango Leaf disease

B. Model Development and Training

This work involves the development of a lightweight CNN model using TensorFlow and Keras to classify mango leaf health into eight distinct classes. The architecture has been designed lightweight so that it runs efficiently on embedded systems like Raspberry Pi. The model processes RGB images resized to 224x224 pixels. The extracted features are then processed by a stack of convolutional layers joined with a dropout layer to avoid overfitting. In this paper, we used 80% of the dataset for training and the remaining 20% for validation. We have used the Adam optimizer and categorical cross-entropy loss function. Training was done for 50 epochs. The results were promising. The model achieved a training accuracy of 94.8% and a validation accuracy of 90.2%. The closer gap between the two accuracies shows that the model works well

with the new data also. On a real-time test on the Raspberry Pi 4, the model took approximately 1.8 seconds to process an image, which is fast enough for practical use in the field.

C. Hardware Integration and System Design

The machine integrates a Raspberry Pi 4, a camera module (Logitech C922 Pro webcam), and an Arduino microcontroller. The Raspberry Pi processes video frames in real time the use of the trained version. When a diseased leaf is detected with a confidence level above the threshold, the Raspberry Pi communicates with the Arduino through a serial connection. The Arduino activates a 12V water pump through a relay module to spray the affected area the usage of a centered sprinkler device.

The approach adopted on this project combines computer vision and hardware automation to offer practical solution for mango tree leaf disease detection. By combining visualization and components, the project describes a technique for agricultural problems, providing valuable insights for the development of agriculture systems.

V. SYSTEM DESIGN

A. Component List

The manufacturing costs of a single device are described in Table II. It includes the basic components needed to connect the proposed device. Please note that these prices are approximate and may vary depending on factors such as supplier, order size and location.

B. Schematic Design

The schematic layout shown in Fig. 2 shows the hardware device including detection of mango tree leaf disease detection, connects Raspberry Pi, Arduino microcontroller, USB digicam, relay module, battery, water pump. This system allows real-time monitoring and automatic preventive actions of mango tree leaves sprayed with pesticides or water.

TABLE II. COMPONENT LIST

Components	Price
Raspberry Pi 4	\$142.79
SanDisk Extreme 128GB 190mbps microSDXC UHS-I Memory Card	\$20.06
Servo Motor SG90	\$1.63
MINI DC12V Brushless Pump AD20P	\$6.27
Lipo Battery 2200mAh 11.1V 3S	\$20.90
Logitech C922 Pro Full HD Webcam	\$108.68
360° Automatic Rotating Garden Lawn Sprinkler System	\$4.43
Total Cost	\$304.76

At the middle of the device, the Raspberry Pi serves as the principle processing unit. It runs image processing algorithms and interfaces with a related USB digicam to research visual statistics. The USB digicam captures high-resolution pics of mango tree leaves and sends them to the Raspberry Pi through a USB interface. Raspberry Pi uses system recognition or image recognition techniques to identify patterns of those images to detect symptoms of leaf diseases.

The blending of those components offers a green and reliable tool for early detection and manage of foliar sicknesses on mango timber.

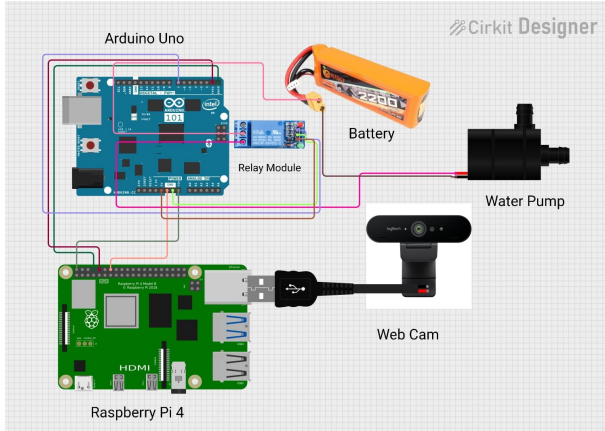


Fig. 2. Schematic Design of Mango Leaf Disease Detection System

C. System Architecture

The diagram of Fig. 3 indicates a “wireless network architecture of a mango tree leaf disease detection machine, the use of imaging strategies to better reveal agriculture. The device starts with a camera capturing photos of mango leaves, which act as records insertion to locate potential diseases. These images are sent to the microcontroller, which acts as an intermediary by preprocessing the data. The server strategies the transmitted data and makes use of picture processing algorithms to come across ailment signs and symptoms on mango leaves.

The processed records is then transferred to cloud storage, ensuring scalability, security and accessibility for long-term

garage and get remote access. A database within the cloud infrastructure organizes statistics for smooth retrieval. Survey consequences are available to end-users through a web-based totally interface, imparting farmers, researchers, agricultural specialists or real-time insights into mango tree health.

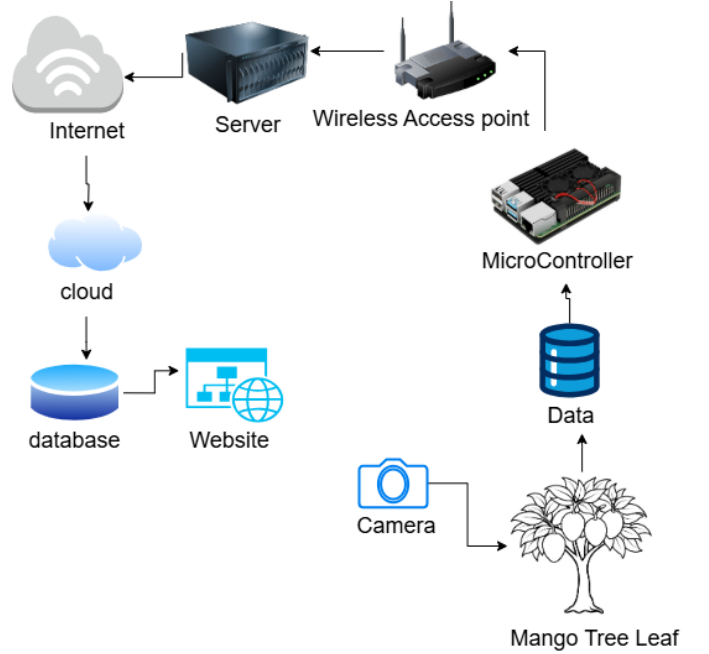


Fig. 3. System Architecture for Mango Leaf Disease Detection System

VI. IMPLEMENTATION

This study offers an integrated framework for real-time mango tree disease detection and automated spray, combining image processing, machine learning and hardware automation. The device has been developed to address the limitations of conventional agricultural practices through presenting flexible, value-effective and automated solutions for disease control.

The hardware configuration for the Mango Leaf Disease Detection Robot is displayed in Fig. 4. The robot comes with a camera module for taking pictures of leaves, a microphone and speaker for voice-based communication, and a 3.5-inch display for live output. A Raspberry Pi acts as the main processor to run the AI model for disease detection, while an Arduino Nano controls the motors via the L298N motor driver. The buck converter regulates the power of the Li-Po batteries (2200mAh and 3000mAh), while the HC-05 Bluetooth module enables wireless communication. The wheel-mounted, self-moving robot can quickly identify anthracnose, powdery mildew, and bacterial blight by scanning mango leaves.

Key features of the proposed solution:

Disease detection using the CNN The core of the machine is a convolutional neural network (CNN) trained on a dataset of mango leaf images. The model correctly classifies leaves as healthful or diseased, with a 90% confidence threshold. Real-time video captured through the Logitech C922 Pro webcam is resized, preprocessed, and fed into CNN for prediction.

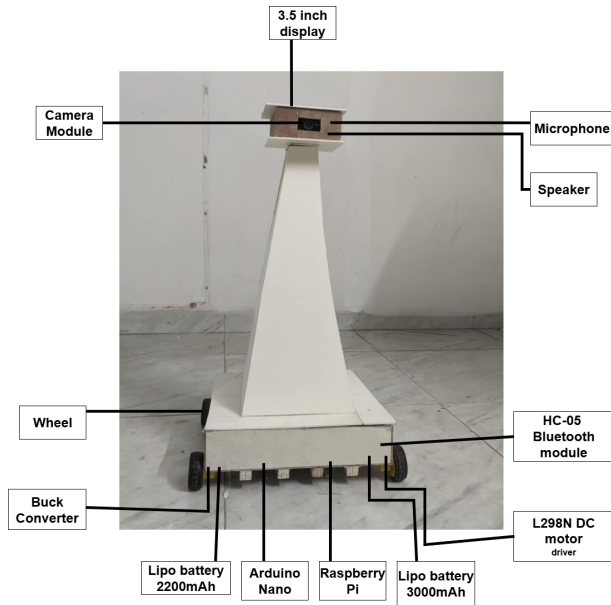


Fig. 4. Test Robot of Mango Leaf Disease Detection System

This ensures well timed detection of diseases which include anthracnose and powdery mildew.

Real-time video analysis:

A Raspberry Pi processes the streaming video and detects the infection. To reduce false positives, Hue-Saturation-Value (HSV) masking is used to analyze only frames with a large percentage of green (indicating foliage).

Automated Spray System:

When a disease is detected with confidence for a sustained period (15 seconds), the Raspberry Pi sends a command to an Arduino microcontroller, which activates a 12V water pump through a relay module. The pump delivers water to the affected areas through a rotating spray system.

IoT integration:

The device uses the connection between the Raspberry Pi and the Arduino to synchronize disease detection. This modular design allows for easy integration and scalability for larger farms.

Durability and usability:

The proposed solution is designed to be easy to use, with minimal technical understanding required for operation and maintenance. Advantages of the proposed solution: 1.High Accuracy: The CNN version achieves reliable predictions through true robust preprocessing and enhancement techniques. 2.Efficient Resource Utilization: Automated Spray ensures efficient use of water, contributing to sustainable agricultural practices. 3.Scalability: Modular system helps in expansion to larger farms or new plants. 4.Real-Time Response: Continuous video analysis to detect spark off and reduce infection. A comparative analysis of the proposed system with currently available industrial products are shown in Table III.

VII. RESULT ANALYSIS

The system's real-time disease detection abilities emphasize the function of machine learning in figuring out plant disease with high accuracy. Our CNN model achieved an accuracy of 98.4 percentage(Photo Mode) and 96.1 percentage(Video Mode), allowing it to reliably distinguish among healthy and diseased mango leaves, together with those affected by anthracnose, powdery mildew shown in Table IV.

A. Disease Classification

The Proposed model's classification performance is summarized up in Table V, where it obtained 94% accuracy on 800 test samples from eight different mango leaf classes. Healthy leaves also performed well (precision = 0.96, recall = 1.00), while Cutting Weevil was classified flawlessly (precision, recall, F1 = 1.00). Sooty Mould was one of the more difficult cases; it had a high precision (0.99) but a low recall (0.80). The model's robustness for multi-class disease detection was confirmed by the overall precision, recall, and F1-score macro and weighted averages of 0.94.

B. Confusion Matrix Analysis

Fig. 5 shows the confusion matrix for the proposed CNN model, highlighting strong performance across most classes. In addition to Gall Midge and Die Back being classified with high reliability, Cutting Weevil, Healthy, and Powdery Mildew all achieved perfect accuracy. Anthracnose was sometimes mistakenly identified as Die Back or Gall Midge, and Bacterial Canker as Die Back, causing some confusion among diseases that shared visual similarities. The main challenge was Sooty Mould, which was sometimes mistaken for Gall Midge, Healthy, or Powdery Mildew, reflecting the difficulty of distinguishing diseases with overlapping visual features.

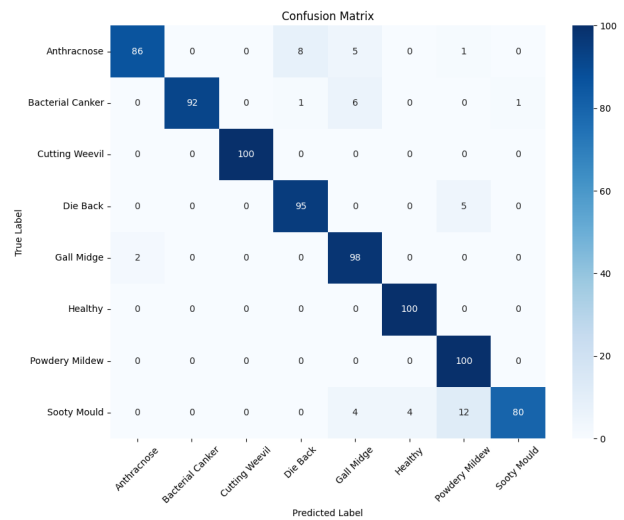


Fig. 5. Confusion Matrix of Convolutional Neural Networking Based Automated Detection of Mango Leaf Diseases

TABLE III. COMPARATIVE ANALYSIS OF THE PROPOSED SYSTEM WITH RESPECT TO CURRENTLY AVAILABLE INDUSTRIAL PRODUCTS

Features	Proposed Device	Croptimus™ by Fermata	xFarm
Approximate Price	\$305	not revealed	Free
Latency	Easy	Easy	Moderate
Real-Time Disease Detection	Yes	Yes	No
Automated Spray System	Yes	No	No
Disease Progress Monitoring	Yes	Yes	No
Scalability	Yes	Yes	Yes
Remote Monitoring	No	Mobile App	Website
Offline Functionality	Yes	No	No
Energy Efficiency	Yes	Yes	No
Data Logging	No	Yes	Yes
Hardware Status Monitoring	Yes	No	No
Ease of Installation	Easy	Easy	Moderate

TABLE IV. PLANT DISEASE DETECTION PERFORMANCE

Test Mode	Accuracy	Avg Time	Notes
Photo Mode	98.4%	~5 sec	Dependent on image quality
Video Mode	96.1%	~10–12 sec	Sensitive to motion and lighting

TABLE V. MANGO LEAF DISEASE CLASSIFICATION REPORT

Class	Precision	Recall	F1-Score	Support
Anthrachnose	0.98	0.86	0.91	100
Bacterial Canker	1.00	0.92	0.96	100
Cutting Weevil	1.00	1.00	1.00	100
Die Back	0.91	0.95	0.93	100
Gall Midge	0.87	0.98	0.92	100
Healthy	0.96	1.00	0.98	100
Powdery Mildew	0.85	1.00	0.92	100
Sooty Mould	0.99	0.80	0.88	100
Accuracy		0.94		800
Macro Avg	0.94	0.94	0.94	800
Weighted Avg	0.94	0.94	0.94	800

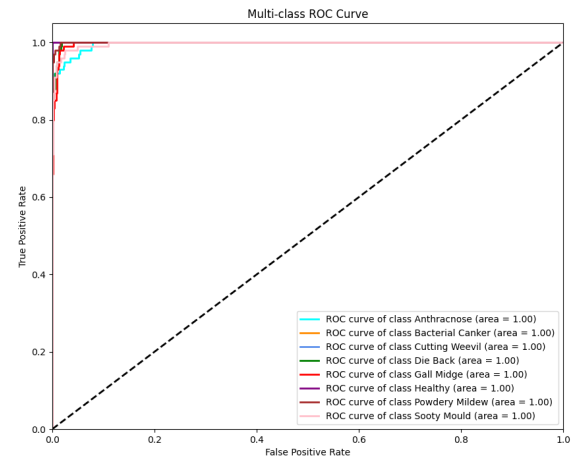


Fig. 6. ROC Curve of Convolutional Neural Networking Based Automated Detection of Mango Leaf Diseases

C. ROC Curve Analysis

Fig. 6 shows the multi-class ROC curves for all disease categories. With all classes achieving an AUC of 1.00, the model exhibits almost perfect discriminative performance, demonstrating a high true positive rate and few false positives across Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Healthy, Powdery Mildew, and Sooty Mould.

D. Importance

This area emphasizes the fundamental importance of robustness in real-world operations, emphasizing that it assures that automation systems remain environmentally reliable in many cases

Broad implications for precision agriculture The high correlation between the accuracy of disease detection and the normal operation of focused Spray systems suggests that combining AI based detection with complete automation based should effectively prevent agricultural yields. This approach

promotes sustainable agricultural practices by enabling precise, disease-focused interventions.

VIII. CONCLUSION AND FUTURE WORK

A. Conclusion

This study succeeded in real-time mango tree disease detection and spray system, combining image processing, machine learning and automation Leveraging a convolutional neural network (CNN) trained on data sets on the system proved to be highly accurate in detecting diseases such as anthracnose and powdery mildew. The system, camera module and Arduino microcontroller, to enable seamless operational communication and efficient hardware control for Automated spray systems. Testing in managed environments established the device's robustness, reliability and potential to function under diverse conditions, and supplied a sustainable and realistic solution for precision agriculture

The challenge addressed key challenges in agriculture through real-time tracking and intervention, reducing water

wastage and reducing the spread of disease. Particularly in the regions of shortage resource and specific agricultural challenges which includes Bangladesh, it is highlighting the capability of integrating advanced technologies to help farmers.

B. Future Work

Future work focuses on extending the data set to contain multiple disease types and frequent images of single environmental conditions to increase sampling generalizability, while the inclusion of real-time domain data enables the tool to adapt to dynamic agricultural conditions. Field applications of the device in real mango orchards will provide insight into its long-term performance, reliability and practicality, and collaboration with local farmers and agricultural experts will help streamline the program for better implementation. Adapting the system to detect diseases in other crops would broaden its applicability and make it a viable tool in precision agriculture, while cost effectiveness can be improved by using a modular system for multi-cropping operations. Sustainability is enhanced by using renewable energy sources such as solar panels to power the system, and energy generation in remote or off-grid areas can support system sustainability. In addition to disease detection, new advanced agricultural support technologies should integrate intelligent spray systems to estimate soil moisture levels. By addressing these areas, the tool can be a robust, scalable and sustainable machine that contributes to the broader dream of precision farming, bridging the gap between industrialization and agriculture to meet global food security and environmental challenges.

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